11 (a) Find the value of $\sqrt[4]{81 \times 10^8}$

2 marks total
Fourth roots 3 and 100
300

$\sqrt[4]{(81 \times 10^8)} = \sqrt[4]{81} \times \sqrt[4]{10^8}$

$= 3 \times 10^2 = 300.$

.....

(2)

(b) Find the value of $64^{-\frac{1}{2}}$

2 marks total
Square root then reciprocal
1/8

64^(-1/2) = 1/√64 = 1/8.

.....

(2)

(c) Write $\frac{3^n}{9^{n-1}}$ as a power of 3

9^(n-1) = (3²)^(n-1) = 3^[2(n-1)].

3^n / 9^(n-1) = 3^[n-2(n-1)] = 3^(2-n).

2 marks total
Write base 9 as 3^2
Subtract exponents

.....

(2)

(Total for Question 11 is 6 marks)

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